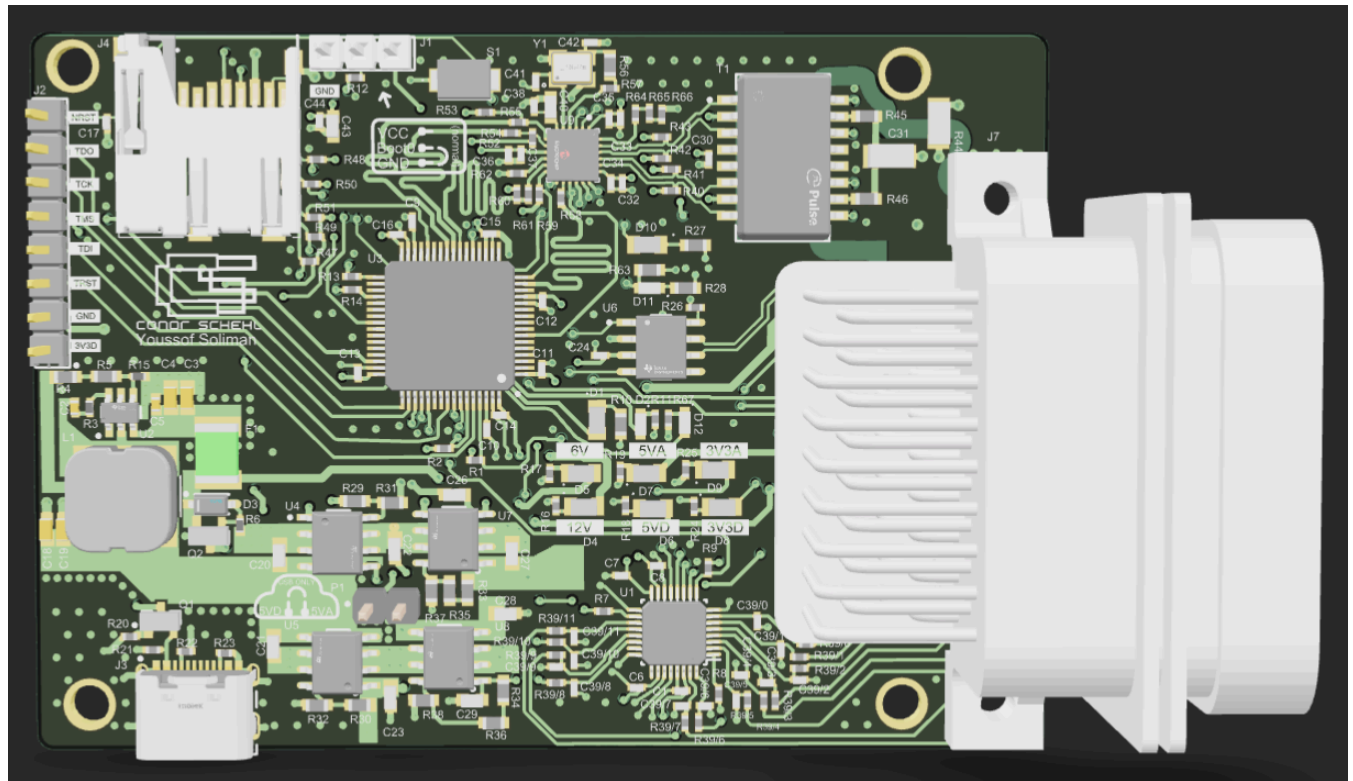


Analog to Digital Converter Data Acquisition Board

-A custom data acquisition board for converting analog signals to digital signals with high precision and low noise



The Objective

In the past, we have struggled with getting readable data from the various analog sensors around the formula racecar. This board was designed to acquire and digitize analog telemetry at a high resolution to be saved and sent at a high speed to the Vehicle Control Unit (VCU). The aim is to fix previous issues surrounding signal integrity and lower resolution analog to digital conversions that we have dealt with in years prior. Midway into development, the scope of this board increased to have it also act as a small power distribution unit to reduce harness complexity.

System Architecture

The architecture is based on an STM32 microcontroller and a 16bit ADC, the ADS114SO8B. This ADC was chosen to optimize the Bill of Materials cost while providing all of the necessary features to alleviate previous complaints. This ADC comes with a fast enough sampling speed and an internal Programmable Gain Amplifier (PGA) to increase flexibility, variability, and reliability, as different types of analog inputs go into it. Communication with the rest of the car is driven through a 100BASE-TX Ethernet PHY using RMII with two MDI channels. Because RJ45 connectors lack the robustness required for automotive

environments, the channels are routed through our standard AMPSEAL connector.

Design Rationale & Challenges

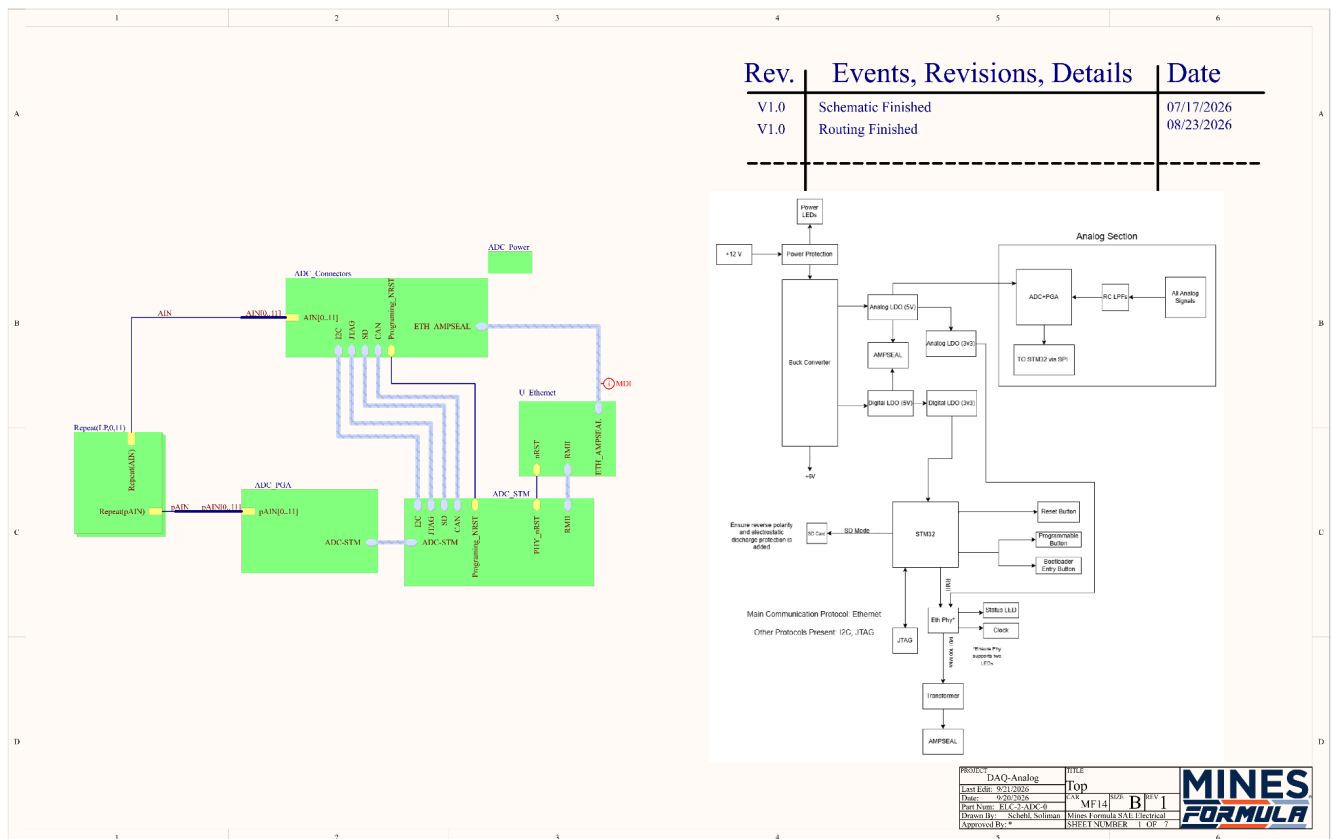
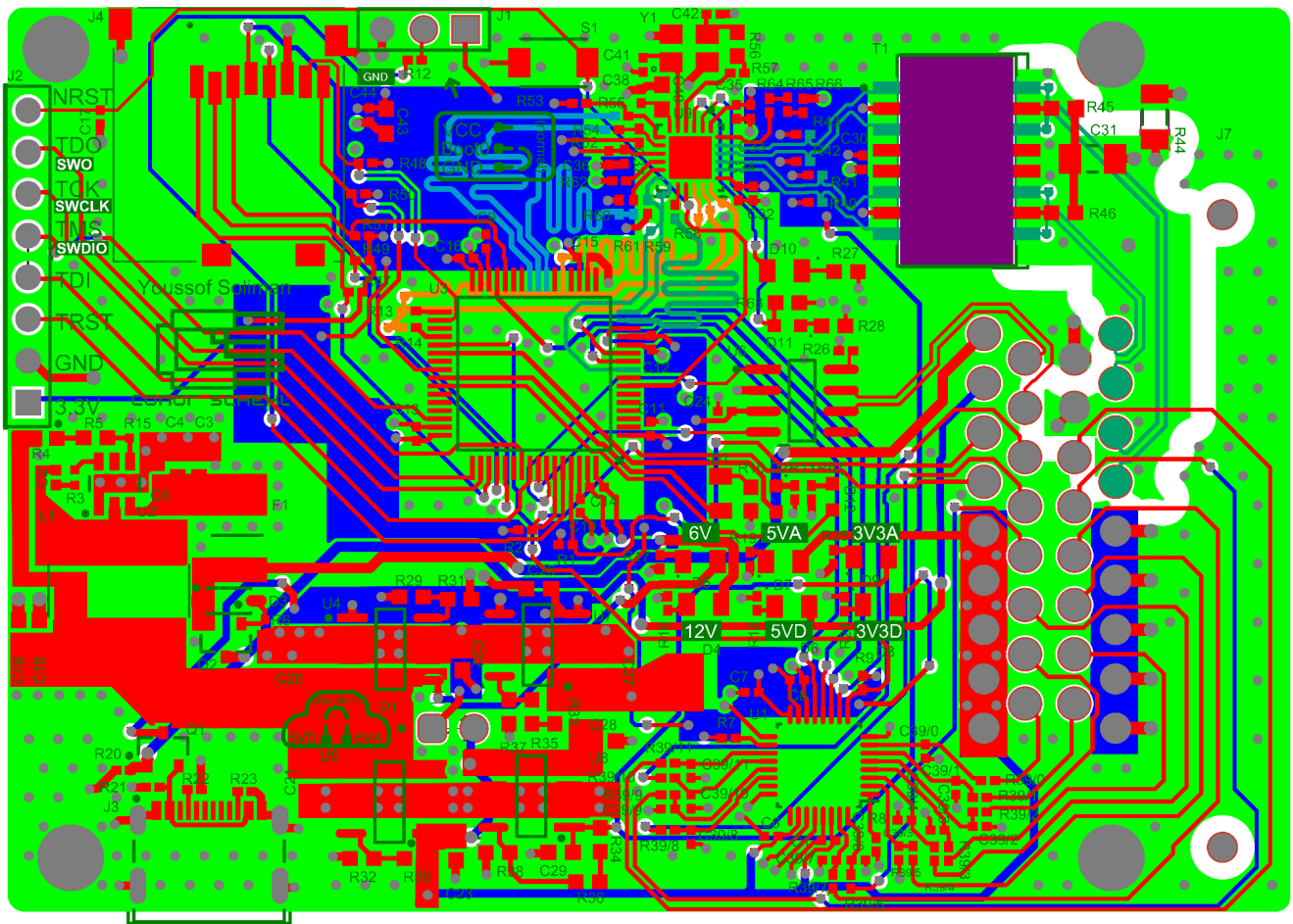
Signal integrity required strict physical separation of the analog and digital sections of the PCB. Instead of physically splitting the ground plane and causing unpredictable return paths along with radiation emissions issues, one solid ground plane is used with careful routing of components over the top. The analog and digital power domains were isolated using the discrete Low-Dropout Regulators (LDOs). The traces that cross the boundary are only the SPI lines that communicate with the STM32 and the headers used with jumper wire in specific bench testing scenarios. By keeping few traces crossing the boundary analog inputs were able to get to their respective low-pass filters without risk of digital crosstalk.

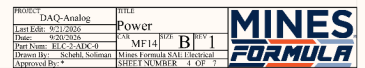
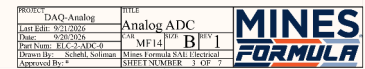
Routing high-speed Ethernet differential pairs through an AMPSEAL connector instead of a standard RJ45 introduced its own set of challenges. Ethernet requires specific magnetics and length matching when going through long transmission wires. The solution to the magnetics issue was to have all hands help to design a discrete magnetics solution with an onboard transformer. Length matching was an issue to be tested later, and it was found that understanding the topology of an AMPSEAL, you can route the differential pairs alongside each other, and the signals are robust enough to maintain reliable communication.

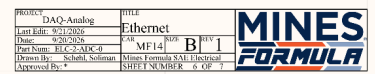
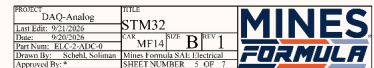
Furthermore, the added requirement to supply external power to various sensors (analog and digital) introduced issues with thermal limitations when dealing with the power budgets. The solution for this was to use an LDO with low thermal resistance and ensure it properly sinks into the copper ground planes underneath with proper thermal relief vias.

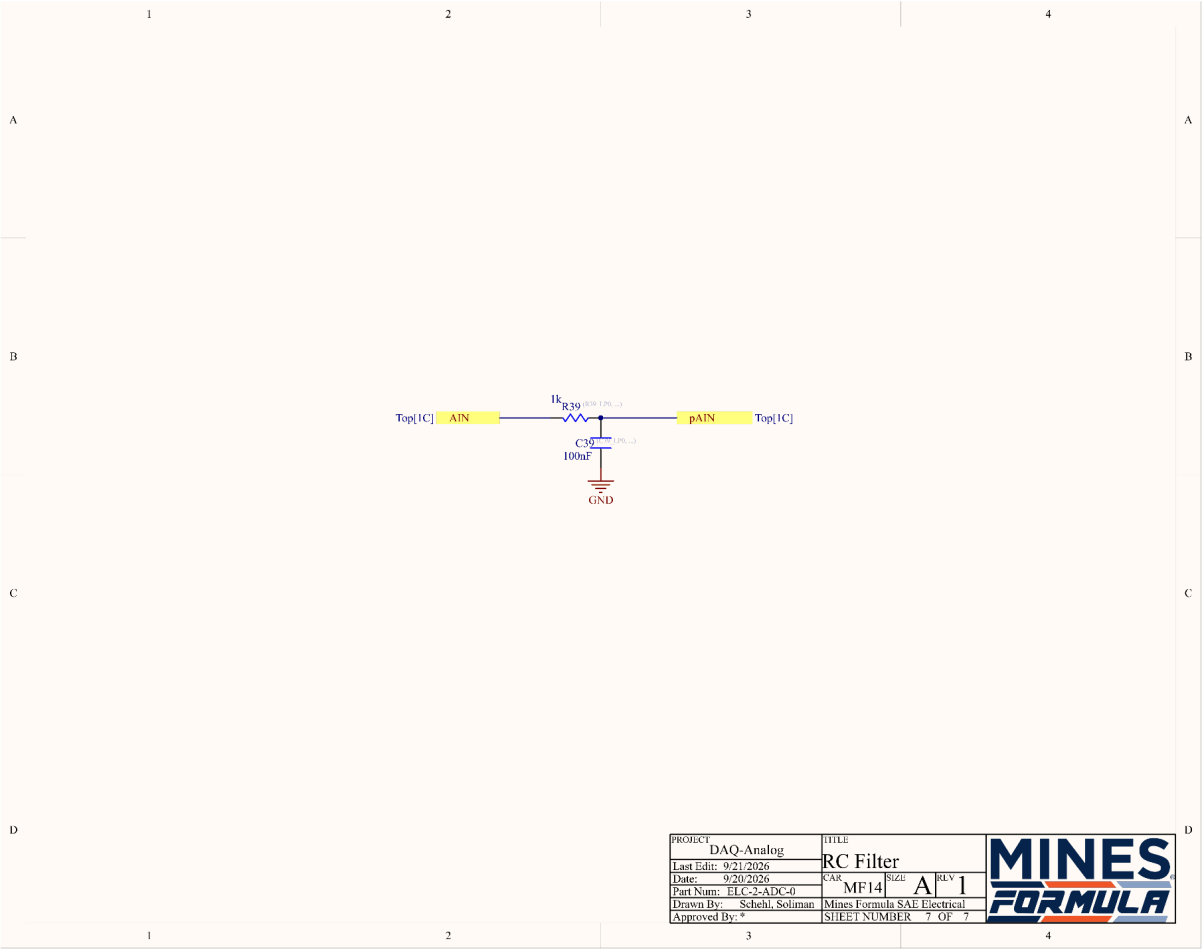
Outcome & Validation

The board successfully managed the thermal dissipation while providing highly stable power sources and voltage references for the analog sensors. The Ethernet solution maintained robust, continuous communication and also allowed for the self-correcting protocols that Ethernet has in place to aid with design and manufacturing challenges despite the non-standard connector.



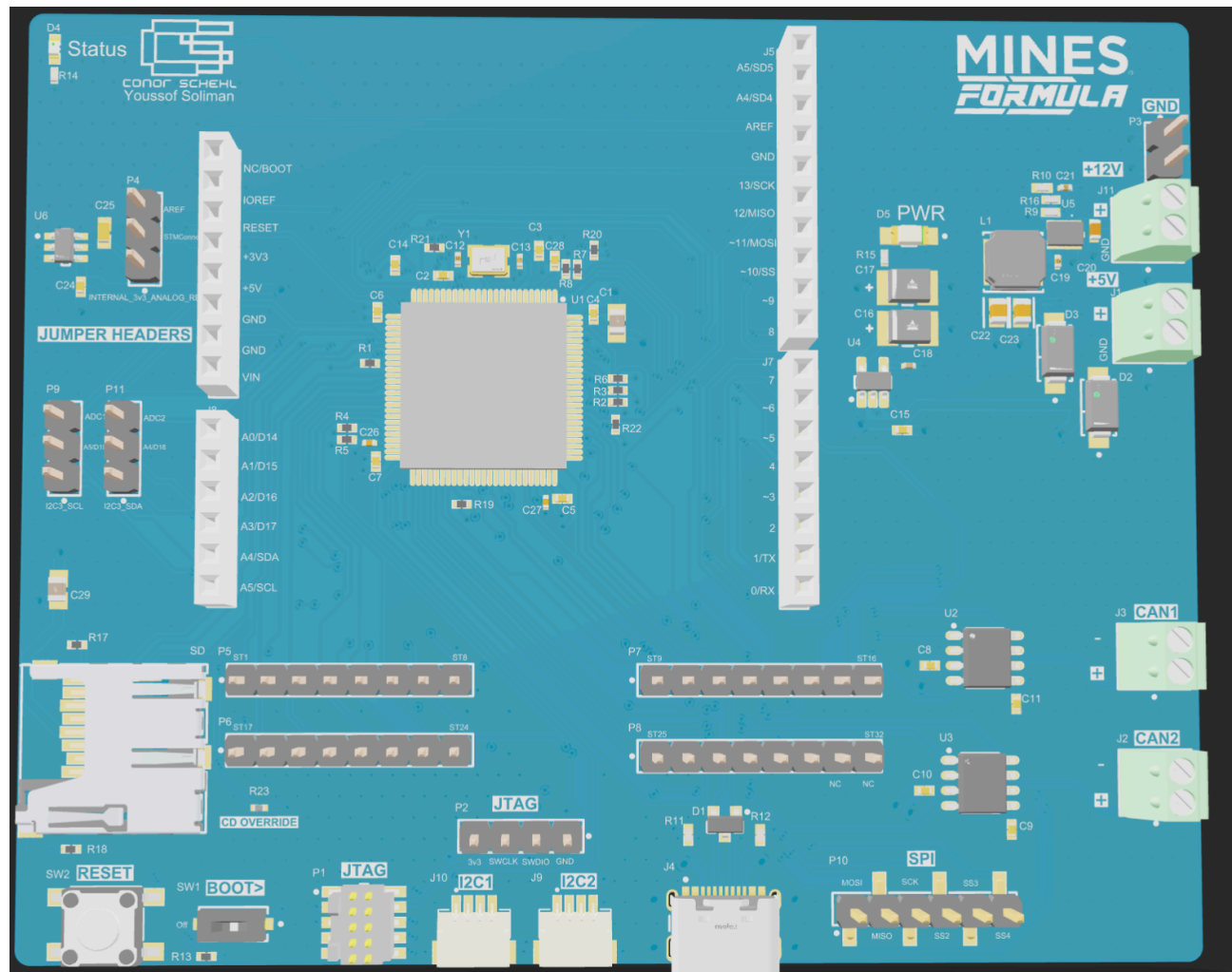






Custom STM32 Development Board

-4-Layer mixed-signal architecture for rapid sensor testing



The Objective

Off-the-shelf development boards are expensive and misaligned with the team's specific STM32 MCU architecture. This custom 4-layer board maximizes GPIO access while bringing out the specific pins to be compatible with Arduino Shields. The cross-compatible nature allows for quick sensor testing, embedded knowledge transfer, and firmware development before the final data acquisition boards are manufactured.

System Architecture

The architecture serves to transition the team from STM32F1 microcontrollers to a higher performance STM32F7 series or similar MCUs with ethernet capability. The upgrade provided the necessary pin density for the current requirements while having plenty of bandwidth for future hardware design improvements, the biggest of which will be Ethernet capability. To maintain budget controls and consolidate the Bill of Materials (BOM), passive

components and power delivery components were standardized using JLCPCB's basic component library, eliminating extended part placement fees while maintaining >80% power conversion efficiency.

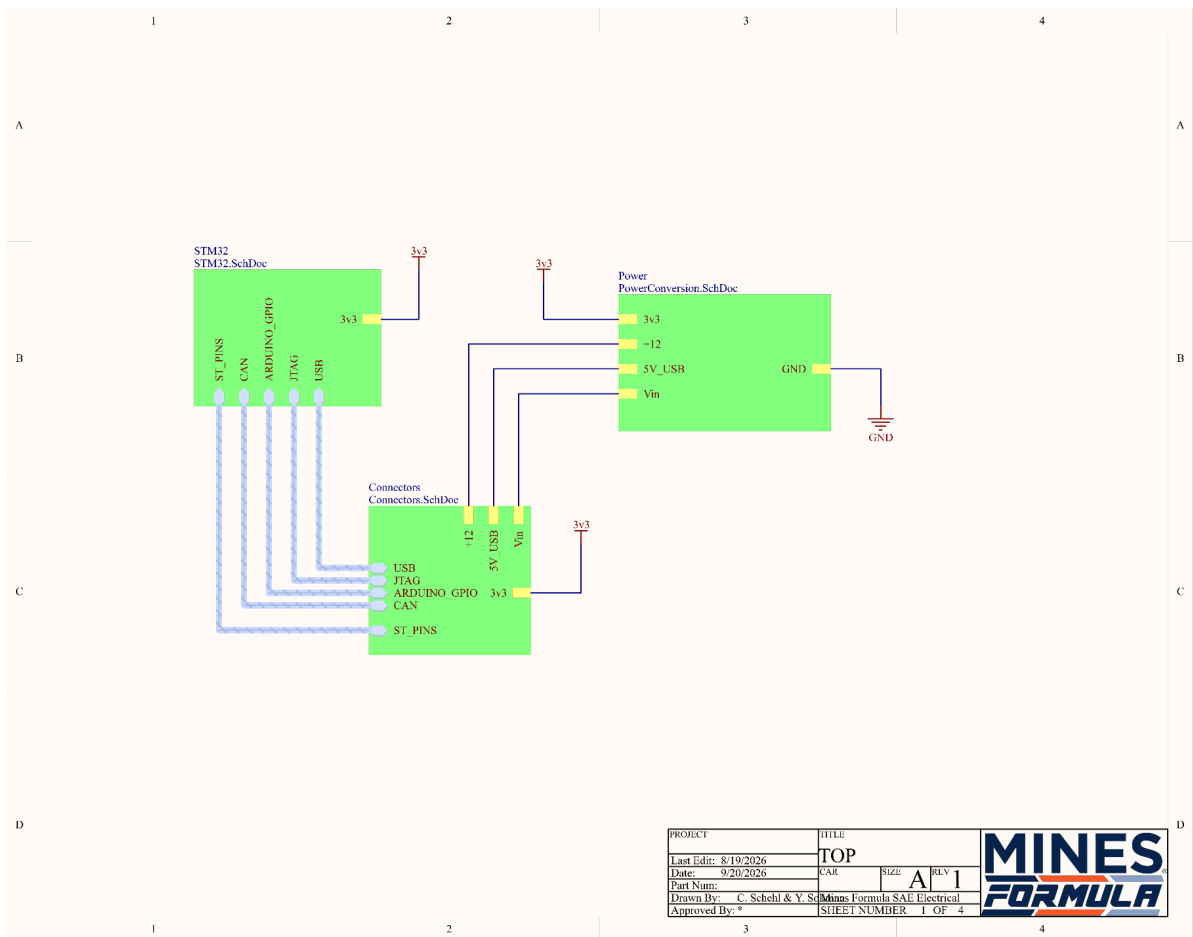
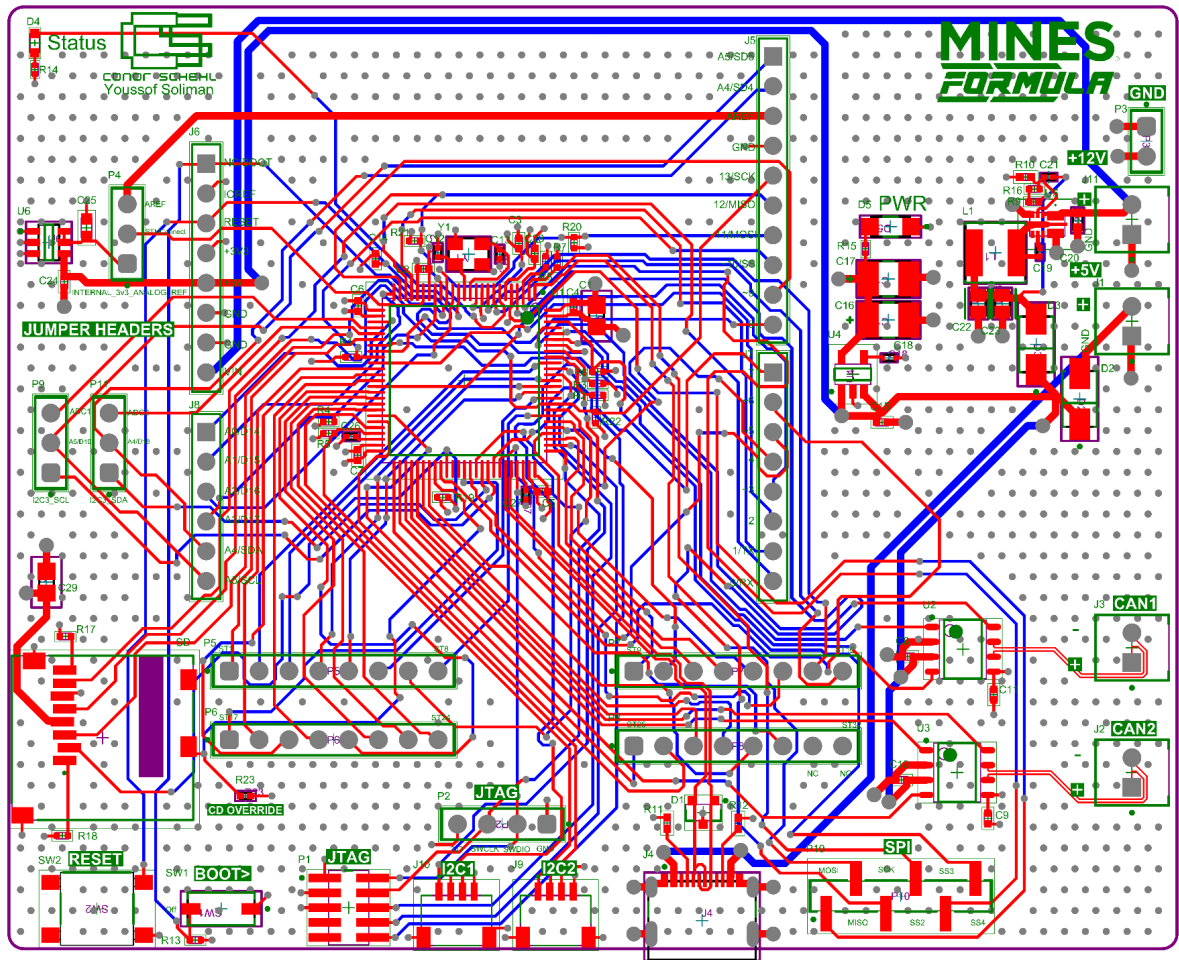
Design Rationale & Challenges

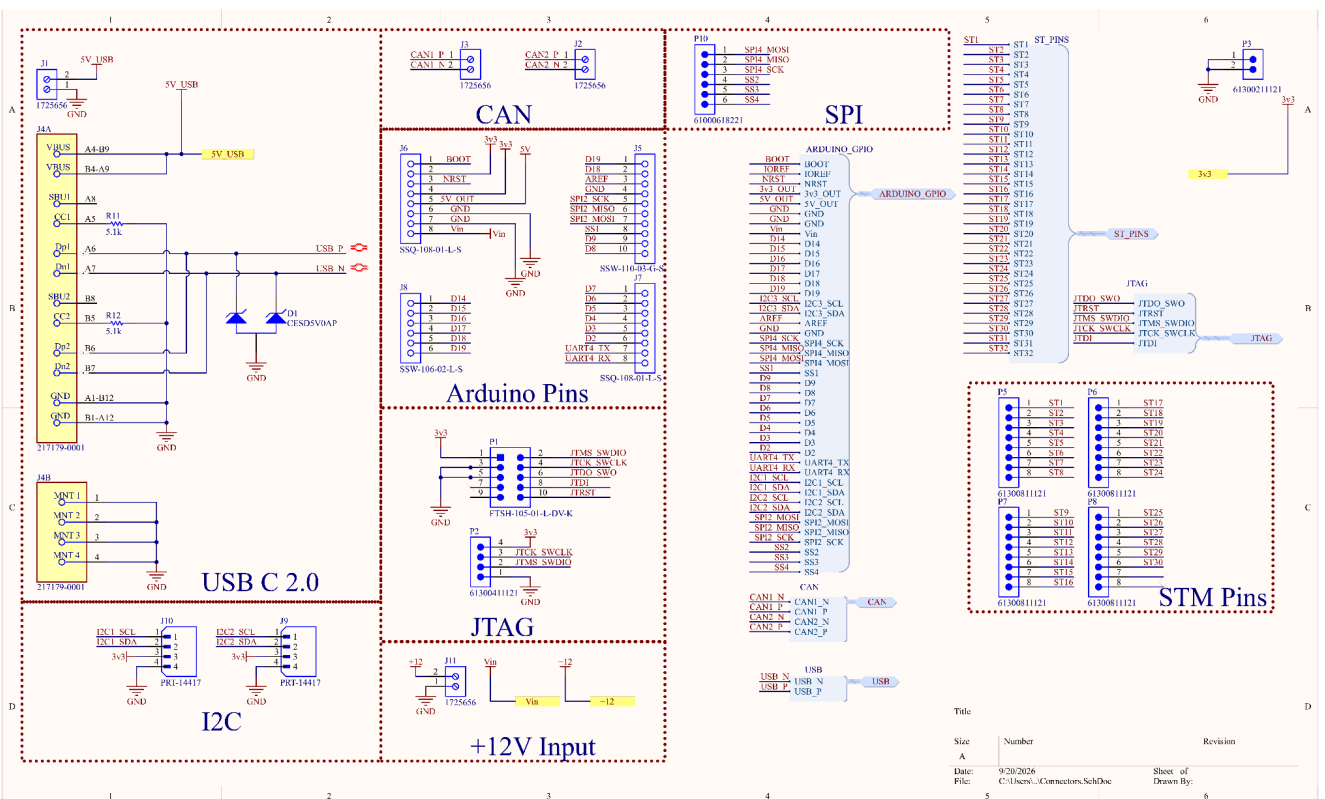
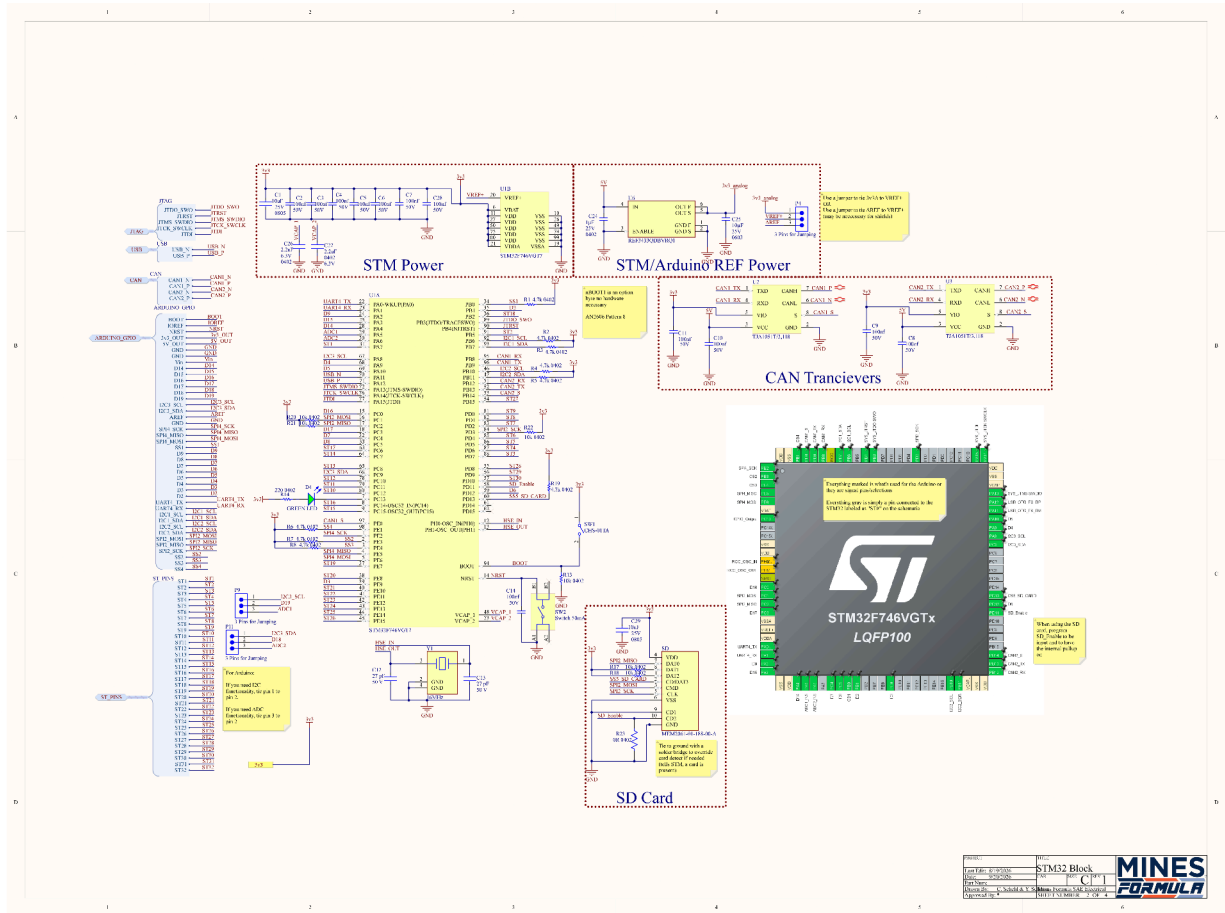
Routing a dense, full-pinout development board required strict adherence to signal integrity principles. To prevent high-frequency ripple from coupling into communication lines, (like USB 2.0 and CAN) differential pairs were strategically routed on the opposite of the MCU, opposite the high-speed oscillator. The impedance-matched USB lines proved spatial and navigation challenges. The trace width was thicker than the CAN traces which meant routing through tight clearances while trying to minimize the length required some strategic rerouting and component placement. Conversely, CAN bus routing was made relatively easier by placing transceivers near the edge of the board.

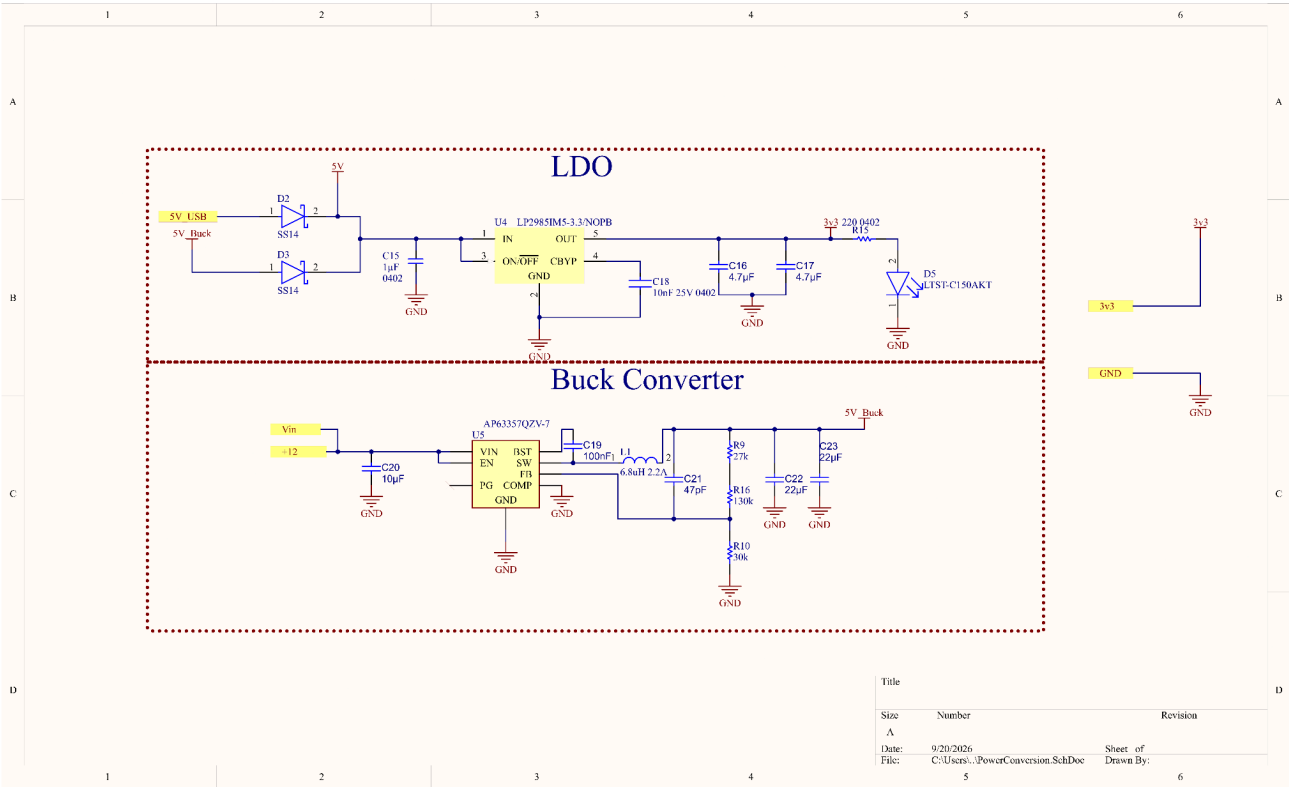
The final challenge came during the board assembly process. A GPIO pin when being placed into the board sheared a copper barrel that's placed in PCB holes for conductivity, and attachment. This shear ripped up the trace that the barrel was attached to. The connection was restored with a hand soldered bodge wire. Everything proceeded to work as expected when testing conductivity signal integrity through the bodge wire.

Outcome & Validation

The board successfully served as the primary hardware testing platform for the electrical and software subsystems. This allowed new members to quickly learn embedded firmware, as well as get hands-on experience programming, testing sensors, and getting comfortable with the team's electrical infrastructure. Clock functionality and thermal stability were validated in multiple environmental states using test code, multimeters, and oscilloscopes, ensuring the MCU maintained reliable timing for data acquisition tasks.



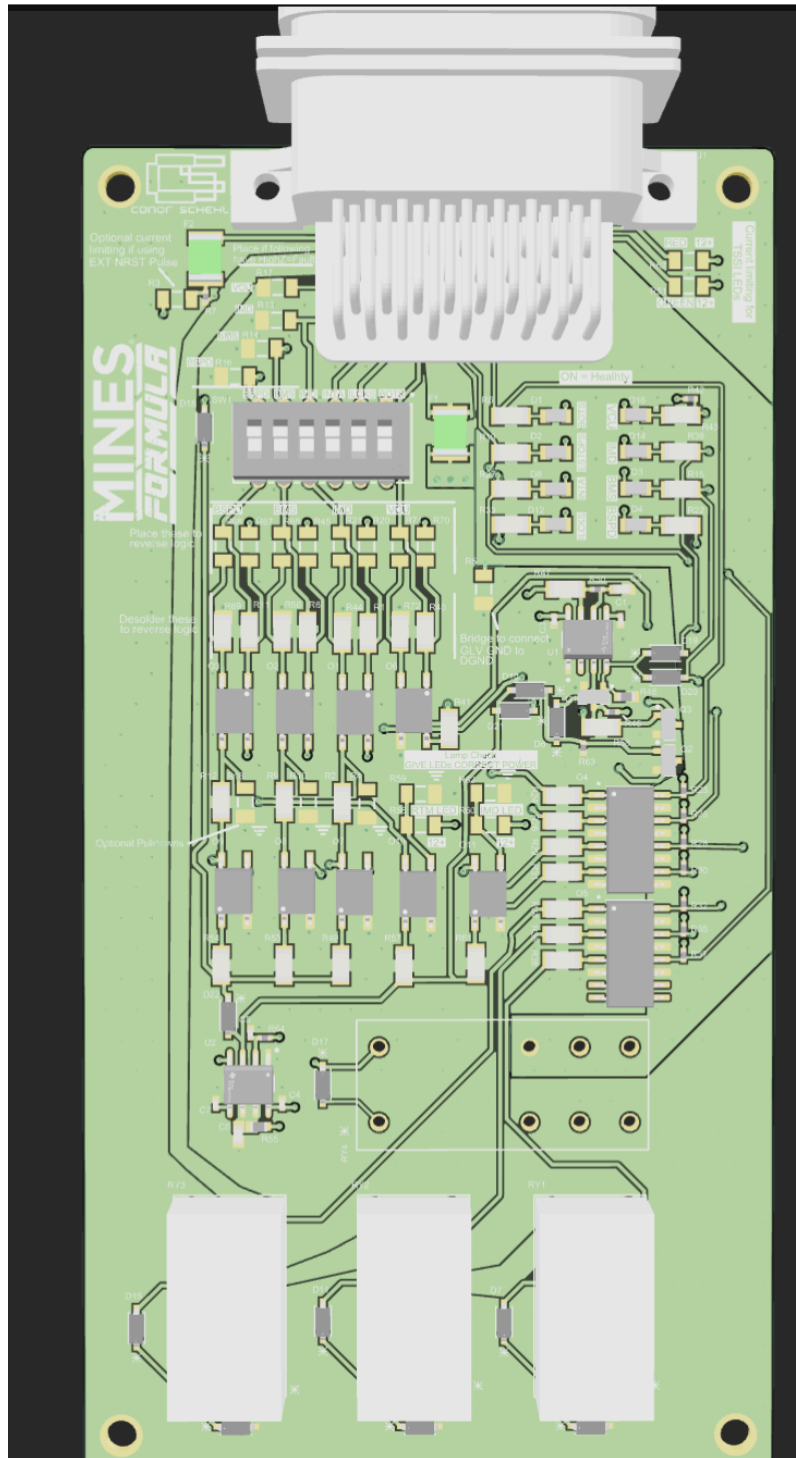




Power and Safety Latching Circuit

- Takes various fault sensor inputs and provides latching functionality disabling the EV high voltage system

Controls status indicator lights for safety, communication, and rapid telemetry gathering



The Objective

The Power and Safety Latching Board is the primary hardware safety barrier for the electric vehicle. Not only is it designed to physically latch and isolate the Battery Management System (BMS), Insulation Monitoring Device (IMD), and Brake System Plausibility Device (BSPD) fault conditions, but also the scope was eventually expanded to include control systems for the car's high-voltage status indicators as well as execute a startup lamp check sequence. Finally, it was deemed necessary to have an avenue for software from the Vehicle Control Unit (VCU) to shut down the car, and the best interface for software shutdown was on this Latching Board.

System Architecture

The core latching/switching logic relies on Omron G6B-22-14P Normally Open (NO), double-pole relays. The NO configuration satisfies the strict competition rules, while containing two poles allows the safety signal to pass through without interacting with the fault signals. CYTLP127(TP) "Darlington" optocouplers with BJT outputs were used for voltage shifting which provided the necessary current to drive the relay coils. Additionally, TLP291-4 optocouplers transmit the status of each fault sensor signal as well as discrete physical automotive fault locations to our digital datalogging and data acquisition (DAQ) board. Later in the design process, a Normally Closed (NC) relay was added so the VCU can trigger a hardware shutdown with software commands.

To manage the visual safety indicators, 555 timers were introduced to the architecture. One 555 timer allows one of the high-voltage status indicators to blink at 3.24Hz, which is in the rules specified frequency range of 2-5Hz. The feedback components added to this circuit were chosen to make it easy to hotswap and change the frequency in the event of a rules change. Another 555 timer was used for the lamp check upon powering the car. As the car powers up, multiple dash and LED indicators around the car must turn on for a set duration to verify the indicators are working. This lamp check sequence also resets and proceeds to allow the Latching Circuit to turn on and function shortly after the car turns on.

Design Rationale & Challenges

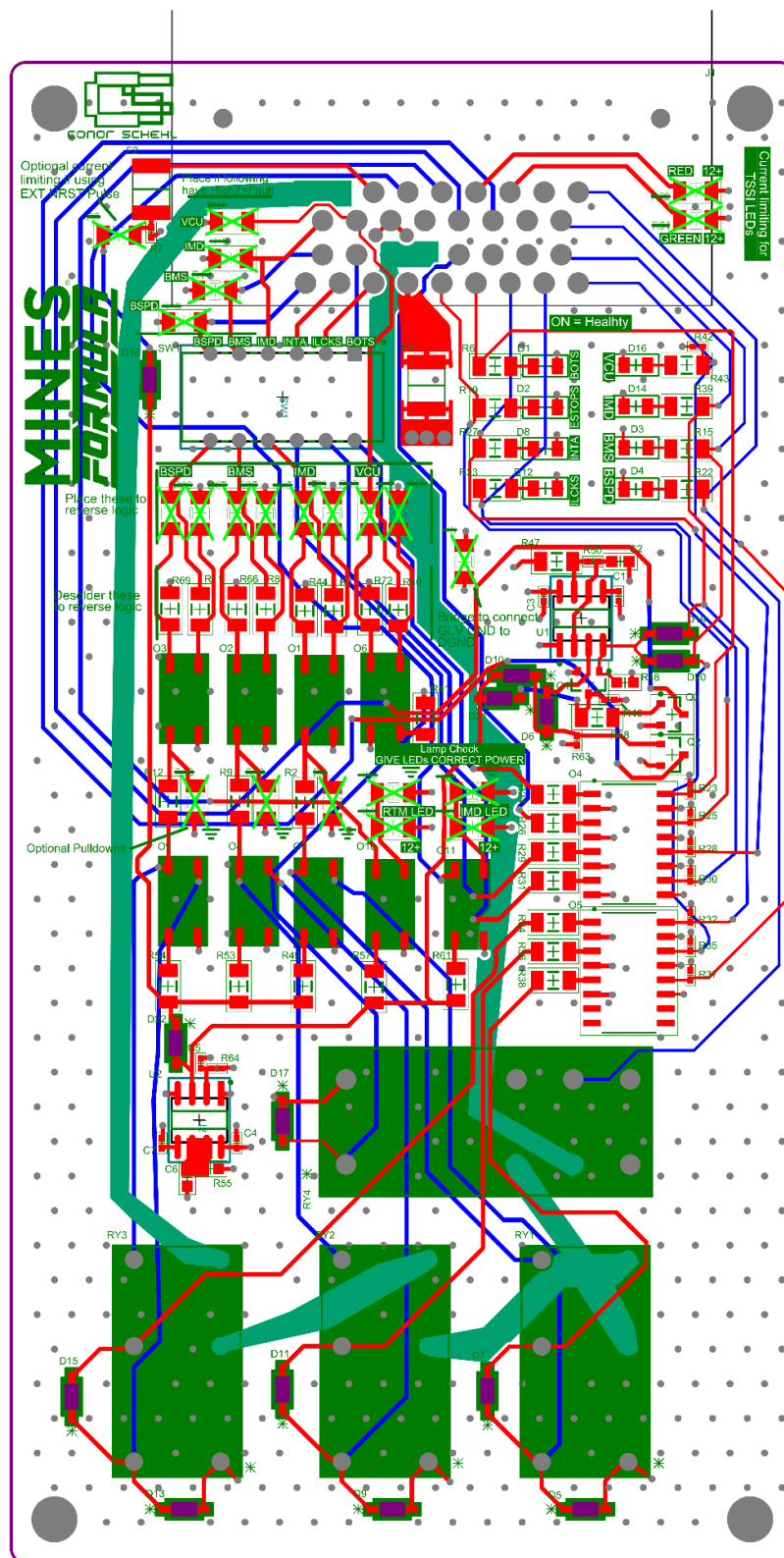
One of the biggest challenges was that this is the first generation of this board, and with that, there are still many unknowns from both a competition side, and our own architecture side. Each fault indicator might have different levels of voltage and current for example. This board is optimized for high-impedance, open-drain faults, with many Do Not Populate (DNP) pads for future proofing and quick configuration if for example it turns out a signal is low-side signaling instead of the expected high-side. Furthermore, back-current protection diodes prevent relay coil inductive kickback from frying the sensor boards, or

even the board traces and other passive components. All this is maintained while the LEDs placed on the board provide immediate visual diagnostics.

Another challenge was integrating all the scope creep additions to the board. Each addition added its own diode ORing logic whether that's different methods of reset or that's adding the lamp check functionality onto the indicator functionality. MOSFET ideal ORing is more stable and reliable, but the board would've gotten way too complex for the scope, as well as add logic that's harder to debug and introduce more places for error. The simpler diode ORing worked by adding the appropriate pull-down networks to ensure the MOSFET gates did not remain floating thus locking conduction when we don't want conduction.

Outcome & Validation

The board successfully isolates the high-voltage system upon receiving the fault signals and sends the faults to the DAQ board while maintaining isolation as verified with the "fault injection" switches. The DNP footprints allowed for easy adapting and evolving of architecture and hardware specifications. The 555 timers allowed for consistent, easily changeable timer logic for the boot sequence and indicators which were easily verified with hand timers or an oscilloscope. Finally the diode ORing maintained the simplicity without sacrificing reliability



RP2040 Thermal Imaging PCB

-Credit-card sized thermal camera with QSPI flash and custom LiPo Battery Management System

The Objective

When I was between hardware projects for Formula, I decided to take on my own project. This project allowed me to explore discrete microcontroller architecture and construct on-board power management. My goal was to build a portable thermal camera without using premade modules or breakout boards. The idea was to get more experience in areas I hadn't worked on with the team as well as make something practical that may assist with hardware testing in the future.

System Architecture

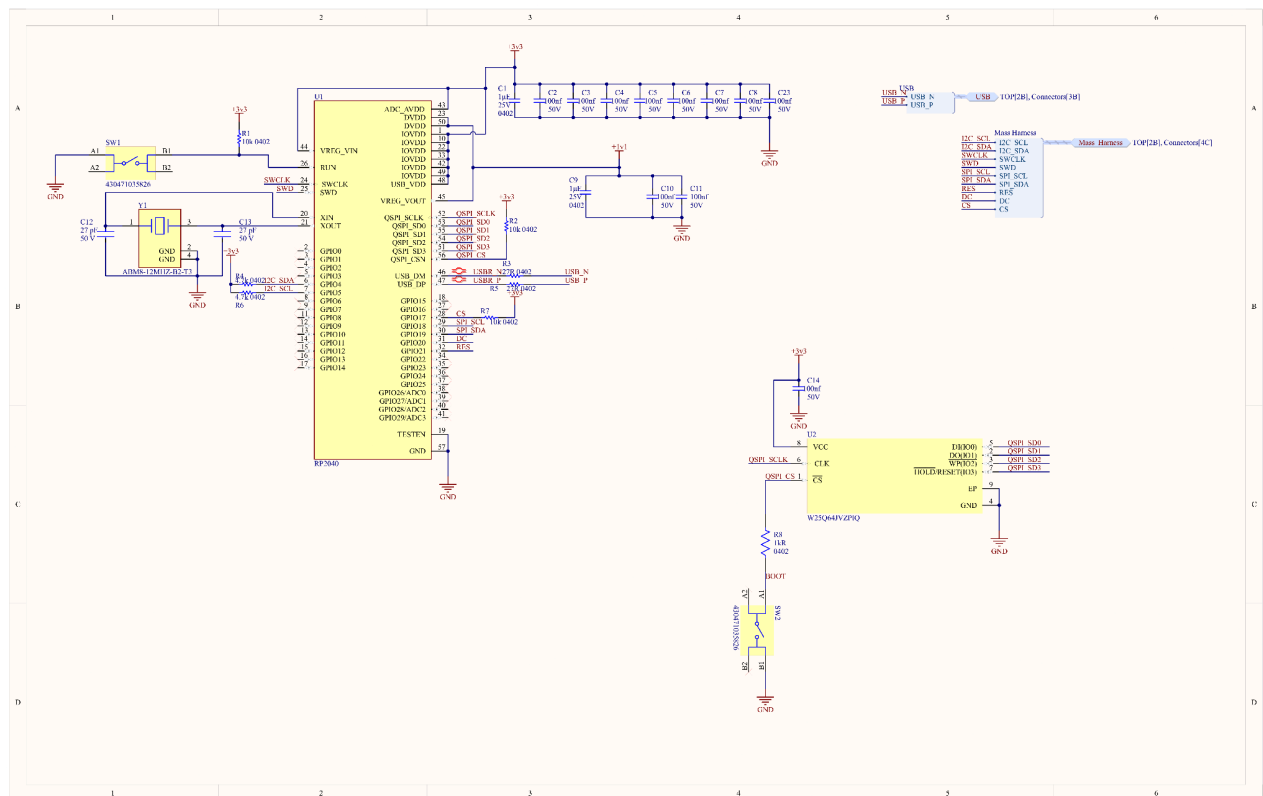
The imager is an MLX90640. The data from the sensor is processed by an RP2040 microcontroller. This MCU lacks internal flash memory, so an external QSPI Flash chip was placed for program storage. To make the camera more portable, a custom LiPo Battery Management System was designed. This system allowed direct power delivery via USB-C which would charge the battery, and power the device.

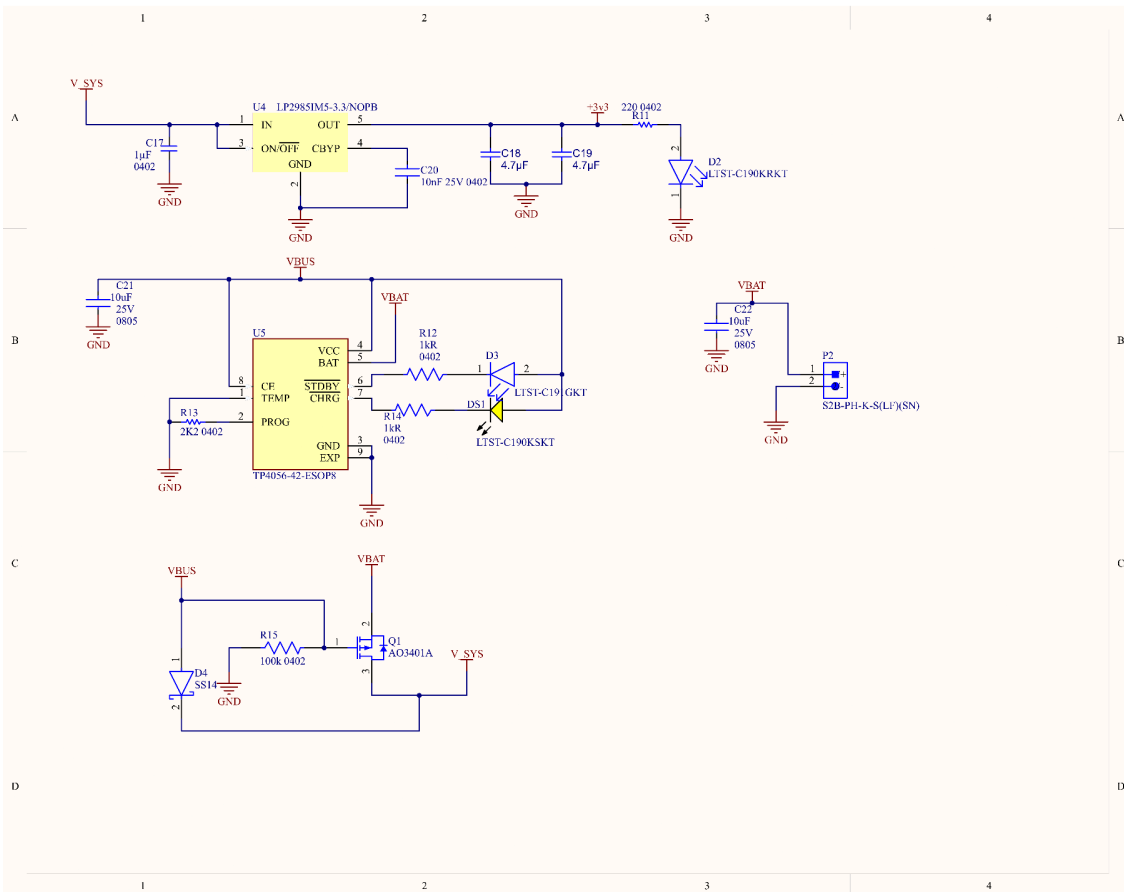
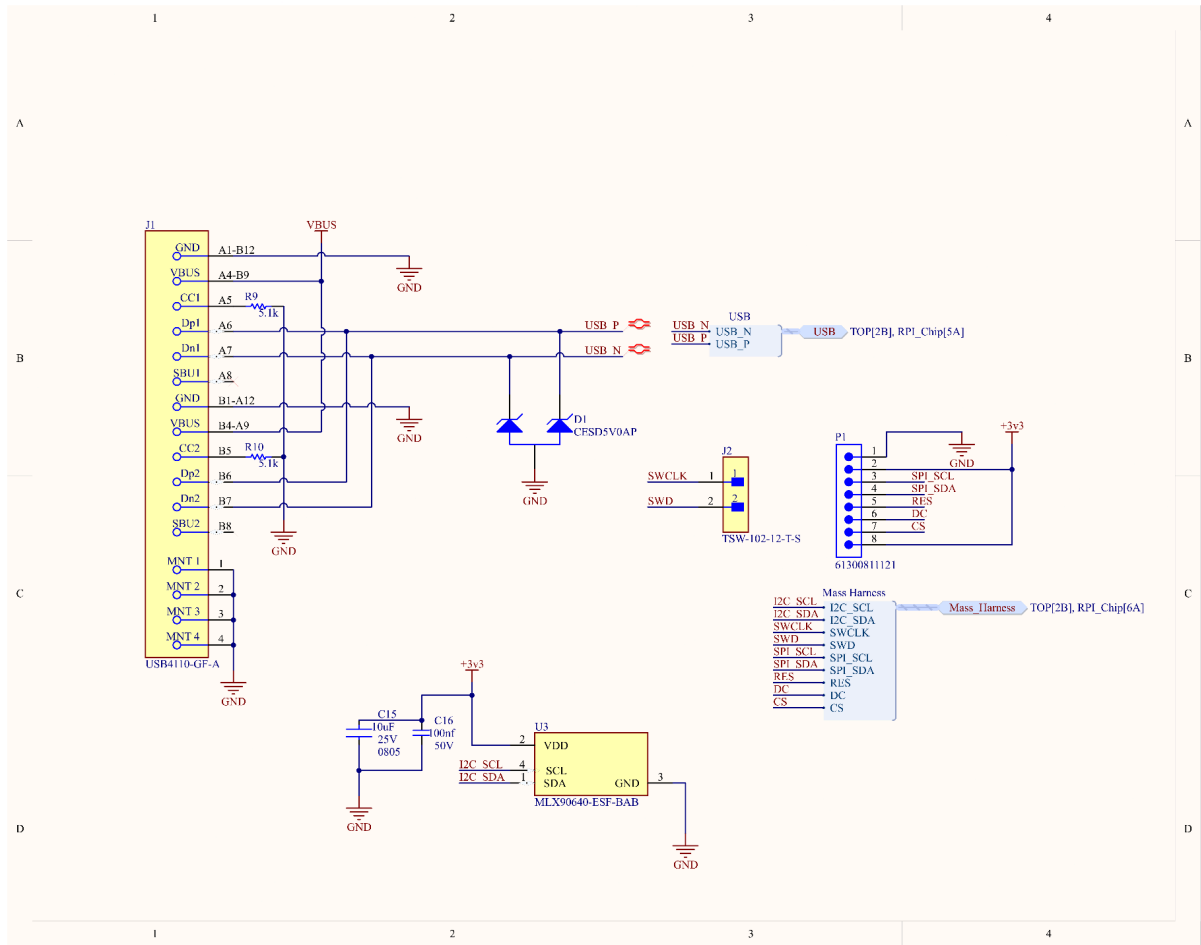
Design Rationale & Challenges

The main challenge was developing the device in such a way to stay mobile, thus, power path efficiency was critical. To prevent the system from constantly charging and draining the LiPo battery, a load-sharing circuit was implemented with a P-Channel MOSFET. This allowed for seamless power transitions between the battery and the USB-C power, keeping the battery healthy increasing its lifetime use. Another challenge was getting the credit-card-sized footprint. This required planning, measuring, and clever component placement and routing to ensure sensitive data lines weren't crossed, everything was stable, and components were thermally sound.

Outcome & Validation

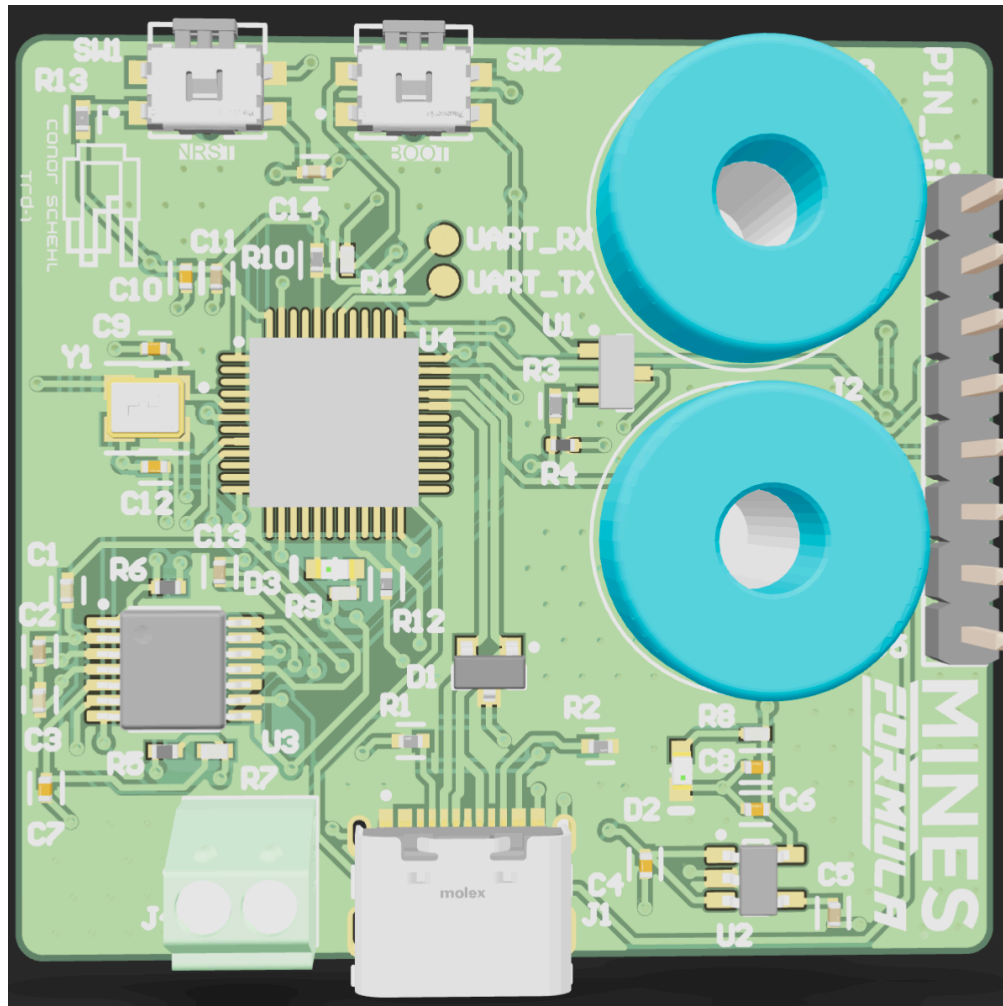
While this board has yet to be ordered and constructed, simulations make me confident in the design, especially around load sharing and power management. Furthermore, I was satisfied with the experience gained with routing a QSPI Flash module as well as engineering my own load sharing circuit.





Dual-Stage Thermocouple DAQ

-My first custom mixed-signal board, used for high-precision temperature measurement



The Objective

On the Mines Formula SAE team, we do everything custom. This includes custom carbon fiber ovens. This 2-layer PCB is a highly accurate analog front-end board for temperature sensing of said ovens. It conditions microvolt-level thermocouple signals, providing a robust, stable voltage to the STM32 MCU to collect, store, and relay the data. This was the first board I made, which was also super exciting!

System Architecture

In the analog domain, the foundation of the architecture is the INA2332 dual-stage instrumentation amplifier. Cold Junction Compensation (CJC) is achieved using an MCP9700 analog temperature sensor IC. The conditioned signal is then digitized by the STM32.

Design Rationale & Challenges

Thermocouple signals are highly susceptible to electromagnetic interference. The biggest challenge for this board despite having a sensitive USB 2.0 differential pair signal, was protecting the fragile thermocouple signal path. Any noise picked up in the path is filtered using a capacitive network before the first amplifier stage.

Instead of using software offsets, the design uses the MCP9700 to get an ambient temperature reference. This voltage is then scaled with a resistor divider and injected into the INA's feedback network, applying CJC at the hardware level. After the first stage, the signal passes through heavy ($200\text{k}\Omega/22\mu\text{F}$) low-pass filtering to eliminate low-frequency noise. Because the filter is at such high impedance, the signal is then amplified through the second stage of the INA in order to prevent the signal from collapsing under the STM32's ADC load. The second stage of the INA acts as an isolation buffer to drive the ADC.

Outcome & Validation

Granted this was my first board, there were lots of mistakes along the way and lots of input and reviews from senior members. Even with all the review, the legacy style INA and schematic proved to be quite confusing and there are improvements on the amplification input stage to be made, especially with the capacitor network and biasing of the amplifier. From my understanding, a broad simplification of the issues is that the input capacitors are incorrectly tied to separate voltage references. This means the ripple and noise is different on each line ruining the common mode rejection ratio. Furthermore, the inputs aren't connected to ground through a pulldown, so the bias return path is floating when the capacitors fully charge. Otherwise, the CJC and filtering designs prove to prevent external EMI from entering the temperature measurement and provide an accurate temperature to the STM32 to be interpreted without complicated software-level smoothing.

